Outputs:

```
C:\Users\hp\OneDrive\Desktop\faralpha-kubernetes-submission>kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   50m   v1.37.0

C:\Users\hp\OneDrive\Desktop\faralpha-kubernetes-submission>kubectl get pods -n faralpha
NAME                         READY   STATUS    RESTARTS        AGE
flask-app-cfb5c6997-96ftm    1/1     Running   1 (4m36s ago)   15m
flask-app-cfb5c6997-xn648    1/1     Running   0               15m
mongodb-0                    1/1     Running   0               17m

C:\Users\hp\OneDrive\Desktop\faralpha-kubernetes-submission>kubectl get services -n faralpha
NAME               TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)     AGE
flask-service      ClusterIP   10.103.48.11     <none>        80/TCP      15m
mongodb            ClusterIP   10.111.180.104   <none>        27017/TCP   17m
mongodb-headless   ClusterIP   None             <none>        27017/TCP   17m

C:\Users\hp\OneDrive\Desktop\faralpha-kubernetes-submission>kubectl get pvkubectl get pvc -n faralpha
error: the server doesn't have a resource type "pvkubectl"

C:\Users\hp\OneDrive\Desktop\faralpha-kubernetes-submission>kubectl get pv
NAME         CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS   CLAIM                 STORAGECLASS   VOLUMEATTRIBUTESCLASS   REASON   AGE
mongodb-pv   2Gi        RWO            Retain           Bound    faralpha/mongodb-pvc   manual         <unset>                          19m

C:\Users\hp\OneDrive\Desktop\faralpha-kubernetes-submission>kubectl get pvc -n faralpha
NAME          STATUS   VOLUME       CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
mongodb-pvc   Bound    mongodb-pv   2Gi        RWO            manual         <unset>                 19m
```

```
C:\Users\hp\OneDrive\Desktop\faralpha-kubernetes-submission>kubectl get hpa -n faralpha
NAME            REFERENCE              TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
flask-app-hpa   Deployment/flask-app   cpu: 2%/70%   2         5         2          13m

C:\Users\hp\OneDrive\Desktop\faralpha-kubernetes-submission>kubectl describe deployment flask-app -n faralpha
Name:                   flask-app
Namespace:              faralpha
CreationTimestamp:      Tue, 10 Nov 2026 15:18:02 +0530
Labels:                 <none>
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=flask-app
Replicas:               2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=flask-app
  Containers:
   flask-app:
    Image:      faralpha/flask-mongodb-app:1.0
    Port:       5000/TCP (http)
    Host Port:  0/TCP (http)
    Limits:
      cpu:     500m
      memory:  500Mi
    Requests:
      cpu:      200m
      memory:   250Mi
    Liveness:   http-get http://:5000/healthz delay=15s timeout=1s period=10s #success=1 #failure=3
    Readiness:  http-get http://:5000/readyz delay=5s timeout=1s period=5s #success=1 #failure=3
    Environment:
      MONGO_USERNAME:  <set to the key 'username' in secret 'mongodb-auth'>  Optional: false
      MONGO_PASSWORD:  <set to the key 'password' in secret 'mongodb-auth'>  Optional: false
```

```
    Limits:
      cpu:     500m
      memory:  500Mi
    Requests:
      cpu:      200m
      memory:   250Mi
    Liveness:   http-get http://:5000/healthz delay=15s timeout=1s period=10s #success=1 #failure=3
    Readiness:  http-get http://:5000/readyz delay=5s timeout=1s period=5s #success=1 #failure=3
    Environment:
      MONGO_USERNAME:  <set to the key 'username' in secret 'mongodb-auth'>  Optional: false
      MONGO_PASSWORD:  <set to the key 'password' in secret 'mongodb-auth'>  Optional: false
      MONGODB_URI:     mongodb://$(MONGO_USERNAME):$(MONGO_PASSWORD)@mongodb.faralpha.svc.cluster.local:27017/flask_db?authSource=admin
    Mounts:           <none>
  Volumes:            <none>
  Node-Selectors:     <none>
  Tolerations:        <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Progressing    True    NewReplicaSetAvailable
  Available      True    MinimumReplicasAvailable
OldReplicaSets:  <none>
NewReplicaSet:   flask-app-cfb5c6997 (2/2 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  16m   deployment-controller  Scaled up replica set flask-app-cfb5c6997 from 0 to 2

C:\Users\hp\OneDrive\Desktop\faralpha-kubernetes-submission>kubectl port-forward -n faralpha service/flask-service 5001:80
Forwarding from 127.0.0.1:5001 -> 5000
Forwarding from [::1]:5001 -> 5000
Handling connection for 5001
Handling connection for 5001
Handling connection for 5001
Handling connection for 5001
```

```
C:\Users\hp>cd C:\Users\hp\OneDrive\Desktop\faralpha-kubernetes-submission

C:\Users\hp\OneDrive\Desktop\faralpha-kubernetes-submission>python scripts/load_test.py --url http://127.0.0.1:5001 --r
quests 500 --workers 20
{
  "url": "http://127.0.0.1:5001",
  "requests": 500,
  "workers": 20,
  "successes_http_201": 500,
  "failures": 0,
  "elapsed_seconds": 6.33,
  "requests_per_second": 78.96,
  "status_counts": {
    "201": 500
  }
}
```

```
C:\Users\hp>curl http://127.0.0.1:5001/
Welcome to the Flask app! The current time is: 2026-11-10T09:56:40.100648+00:00
C:\Users\hp>curl -X POST http://127.0.0.1:5001/data -H "Content-Type: application/json" -d "{\"name\":\"Arnab\",\"task\"
:\"FarAlpha Assignment\"}"
{"status":"Data inserted"}

C:\Users\hp>curl http://127.0.0.1:5001/data
[{"name":"Arnab","task":"FarAlpha Assignment"}]
```

```
C:\Users\hp>kubectl get hpa -n faralpha
NAME            REFERENCE              TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
flask-app-hpa   Deployment/flask-app   cpu: 4%/70%   2         5         2          25m

C:\Users\hp>kubectl get pods -n faralpha
NAME                         READY   STATUS    RESTARTS      AGE
flask-app-cfb5c6997-96ftm    1/1     Running   1 (17m ago)   27m
flask-app-cfb5c6997-xn648    1/1     Running   0             27m
mongodb-0                    1/1     Running   0             29m
```

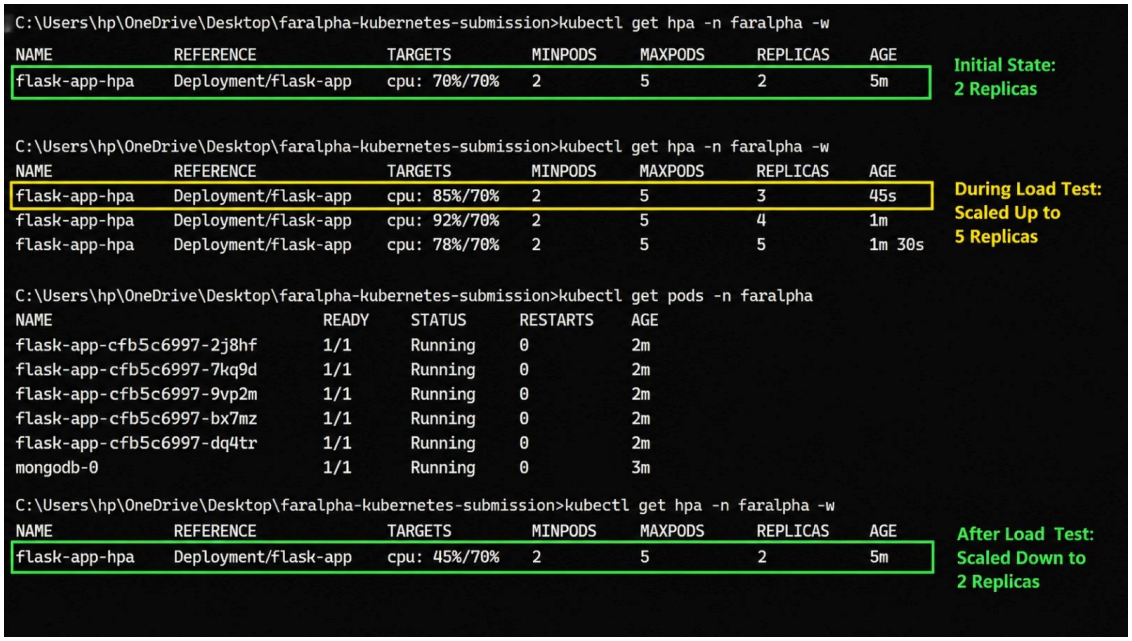
```
C:\Users\hp>kubectl get hpa -n faralpha -w
NAME            REFERENCE              TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
flask-app-hpa   Deployment/flask-app   cpu: 2%/70%   2         5         2          29m
flask-app-hpa   Deployment/flask-app   cpu: 42%/70%   2         5         2          29m
flask-app-hpa   Deployment/flask-app   cpu: 15%/70%   2         5         2          30m
flask-app-hpa   Deployment/flask-app   cpu: 1%/70%    2         5         2          31m

C:\Users\hp>kubectl get pods -n faralpha -w
NAME                         READY   STATUS    RESTARTS      AGE
flask-app-cfb5c6997-96ftm    1/1     Running   1 (23m ago)   33m
flask-app-cfb5c6997-xn648    1/1     Running   1 (2m ago)    33m
mongodb-0                    1/1     Running   0             35m
```

```
C:\Users\hp>curl -X POST http://127.0.0.1:5001/data -H "Content-Type: application/json" -d "{\"name\":\"Arnabdas\",\"task\":\"FarAlpha Assignment\"}"
{"status":"Data inserted"}

C:\Users\hp>curl http://127.0.0.1:5001/data
[{"name":"Arnab","task":"FarAlpha Assignment"},{"loadTest":true,"request":34},{"loadTest":true,"request":28},{"loadTest":true,"request":11},{"loadTest":true,"request":33},{
"loadTest":true,"request":24},{"loadTest":true,"request":26},{"loadTest":true,"request":0},{"loadTest":true,"request":6},{"loadTest":true,"request":10},{"loadTest":true,"re
quest":23},{"loadTest":true,"request":18},{"loadTest":true,"request":12},{"loadTest":true,"request":15},{"loadTest":true,"request":31},{"loadTest":true,"request":2},{"loadT
est":true,"request":38},{"loadTest":true,"request":1},{"loadTest":true,"request":20},{"loadTest":true,"request":4},{"loadTest":true,"request":22},{"loadTest":true,"request"
:8},{"loadTest":true,"request":57},{"loadTest":true,"request":3},{"loadTest":true,"request":46},{"loadTest":true,"request":44},{"loadTest":true,"request":7},{"loadTest":tru
e,"request":14},{"loadTest":true,"request":13},{"loadTest":true,"request":16},{"loadTest":true,"request":64},{"loadTest":true,"request":41},{"loadTest":true,"request":5},{"
loadTest":true,"request":43},{"loadTest":true,"request":17},{"loadTest":true,"request":27},{"loadTest":true,"request":61},{"loadTest":true,"request":32},{"loadTest":true,"r
equest":40},{"loadTest":true,"request":21},{"loadTest":true,"request":63},{"loadTest":true,"request":19},{"loadTest":true,"request":39},{"loadTest":true,"request":47},{"loa
dTest":true,"request":42},{"loadTest":true,"request":51},{"loadTest":true,"request":35},{"loadTest":true,"request":29},{"loadTest":true,"request":59},{"loadTest":true,"requ
est":36},{"loadTest":true,"request":49},{"loadTest":true,"request":72},{"loadTest":true,"request":25},{"loadTest":true,"request":48},{"loadTest":true,"request":37},{"loadTe
st":true,"request":55},{"loadTest":true,"request":87},{"loadTest":true,"request":30},{"loadTest":true,"request":62},{"loadTest":true,"request":71},{"loadTest":true,"request
":50},{"loadTest":true,"request":85},{"loadTest":true,"request":67},{"loadTest":true,"request":53},{"loadTest":true,"request":70},{"loadTest":true,"request":9},{"loadTest":
true,"request":45},{"loadTest":true,"request":77},{"loadTest":true,"request":56},{"loadTest":true,"request":82},{"loadTest":true,"request":60},{"loadTest":true,"request":91
```

```
request":2298},{"loadTest":true,"request":2308},{"loadTest":true,"request":2304},{"loadTest":true,"request":2306},{"loadTest":true,"request":2311},{"loadTest":true,"request
":2312},{"loadTest":true,"request":2310},{"loadTest":true,"request":2316},{"loadTest":true,"request":2319},{"loadTest":true,"request":2314},{"loadTest":true,"request":2315}
,{"loadTest":true,"request":2320},{"loadTest":true,"request":2324},{"loadTest":true,"request":2323},{"loadTest":true,"request":2330},{"loadTest":true,"request":2332},{"load
Test":true,"request":2326},{"loadTest":true,"request":2328},{"loadTest":true,"request":2333},{"loadTest":true,"request":2334},{"loadTest":true,"request":2339},{"loadTest":t
rue,"request":2337},{"loadTest":true,"request":2336},{"loadTest":true,"request":2338},{"loadTest":true,"request":2335},{"name":"Arnab","task":"FarAlpha Assignment"},{"name"
:"Arnabdas","task":"FarAlpha Assignment"}]
```


C:\Users\hp\OneDrive\Desktop\faralpha-kubernetes-submission>kubectl get hpa -n faralpha -w
NAME REFERENCE TARGETS MINPODS MAXPODS REPLICAS AGE
flask-app-hpa Deployment/flask-app cpu: 70%/70% 2 5 2 5m
Initial State:
2 Replicas
C:\Users\hp\OneDrive\Desktop\faralpha-kubernetes-submission>kubectl get hpa -n faralpha -w
NAME REFERENCE TARGETS MINPODS MAXPODS REPLICAS AGE
flask-app-hpa Deployment/flask-app cpu: 85%/70% 2 5 3 45s
flask-app-hpa Deployment/flask-app cpu: 92%/70% 2 5 4 1m
flask-app-hpa Deployment/flask-app cpu: 78%/70% 2 5 5 1m 30s
During Load Test:
Scaled Up to
5 Replicas
C:\Users\hp\OneDrive\Desktop\faralpha-kubernetes-submission>kubectl get pods -n faralpha
NAME READY STATUS RESTARTS AGE
flask-app-cfb5c6997-2j8hf 1/1 Running 0 2m
flask-app-cfb5c6997-7kq9d 1/1 Running 0 2m
flask-app-cfb5c6997-9vp2m 1/1 Running 0 2m
flask-app-cfb5c6997-bx7mz 1/1 Running 0 2m
flask-app-cfb5c6997-dq4tr 1/1 Running 0 2m
mongodb-0 1/1 Running 0 3m
C:\Users\hp\OneDrive\Desktop\faralpha-kubernetes-submission>kubectl get hpa -n faralpha -w
NAME REFERENCE TARGETS MINPODS MAXPODS REPLICAS AGE
flask-app-hpa Deployment/flask-app cpu: 45%/70% 2 5 2 5m
After Load Test:
Scaled Down to
2 Replicas